Infinite Estimates in Logistic Regression

Grinnell College

October 30, 2026

Slide 1 - notes and edits

A Small Logistic Regression Example

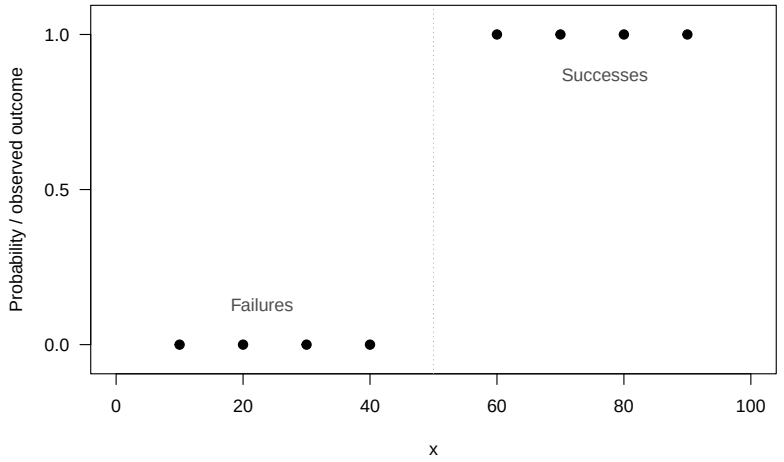
Consider the following eight observations:

```
> x <- c(10, 20, 30, 40, 60, 70, 80, 90)
> y <- c(0, 0, 0, 0, 1, 1, 1, 1)
> toy <- data.frame(x, y)
> toy
   x y
1 10 0
2 20 0
3 30 0
4 40 0
5 60 1
6 70 1
7 80 1
8 90 1
```

Can we predict the outcome perfectly using a cutoff for x?

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Complete Separation



All failures have $x \leq 40$ and all successes have $x \geq 60$. A cutoff at 50 separates the two outcomes.

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The Logistic Curve as the Slope Increases

Recall the fitted probability:

$$\pi(x) = \frac{\exp(\alpha + \beta x)}{1 + \exp(\alpha + \beta x)}.$$

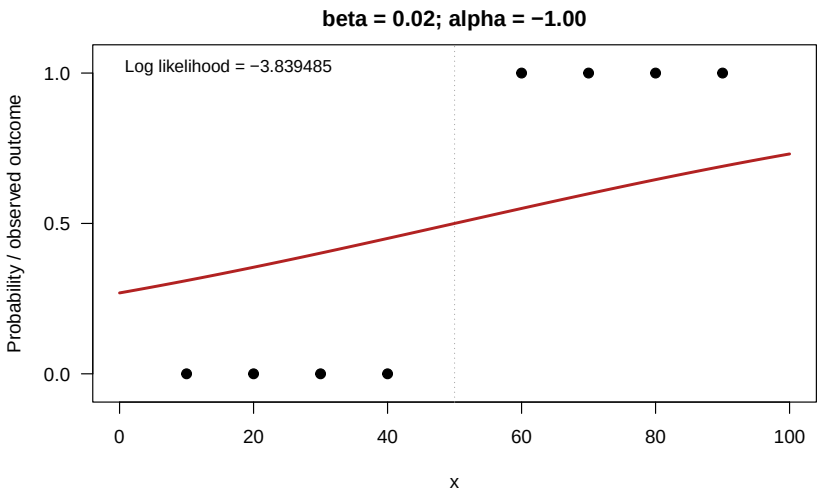
To keep the midpoint at $x = 50$, set $\alpha = -50\beta$:

$$\text{logit}\{\pi(x)\} = \beta(x - 50).$$

- $x = 50$: $\pi(x) = 0.5$ for every value of β.
- $x < 50$: increasing $\beta > 0$ pushes $\pi(x)$ toward 0.
- $x > 50$: increasing $\beta > 0$ pushes $\pi(x)$ toward 1.

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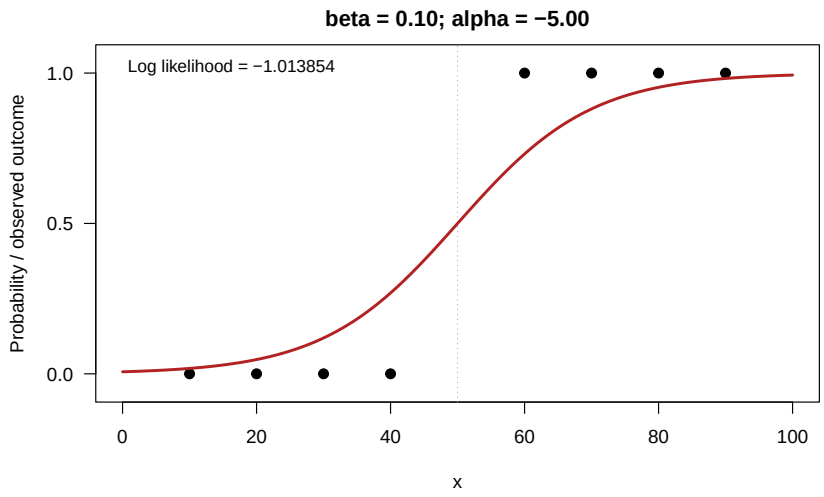
Increasing the Slope: $\beta = 0.02$



The curve assigns moderate probabilities to both outcomes.

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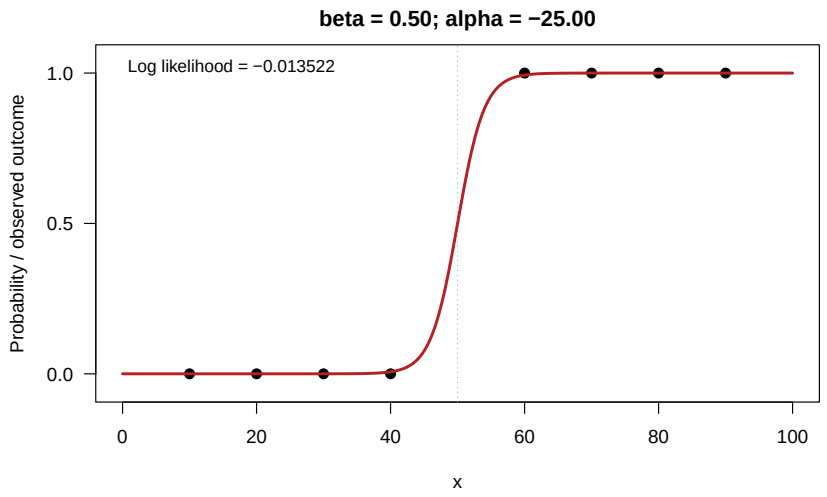
Increasing the Slope: $\beta = 0.10$



Each observed outcome receives a higher probability.

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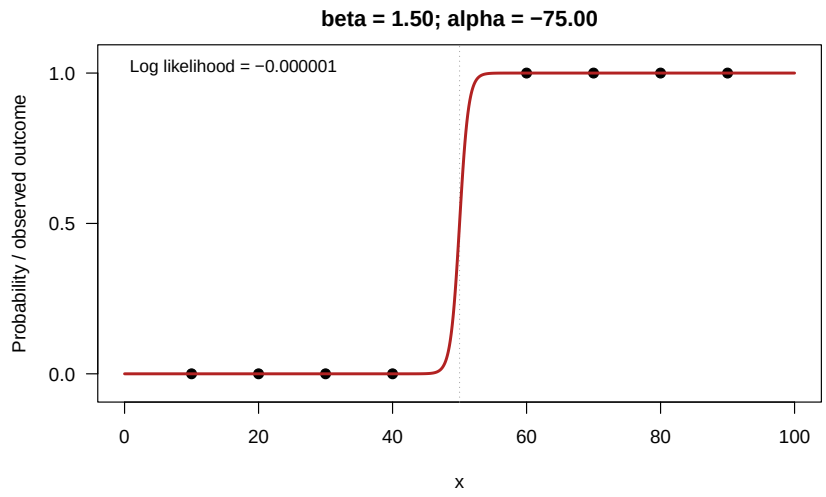
Increasing the Slope: $\beta = 0.50$



Predicted probabilities are now close to the observed zeros and ones.

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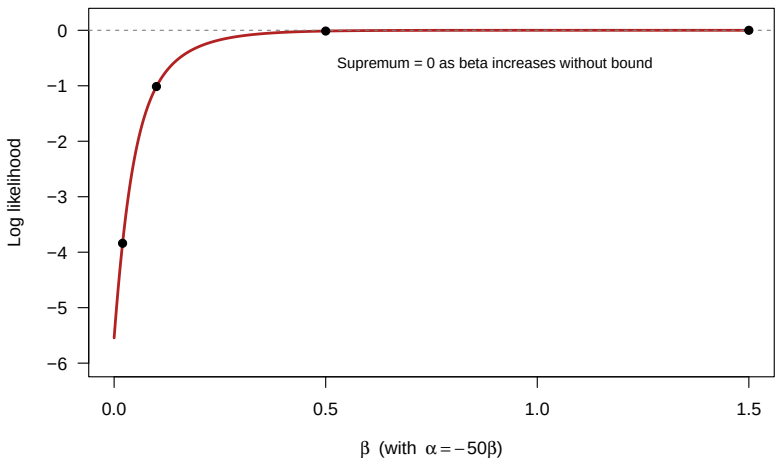
Increasing the Slope: $\beta = 1.50$



The likelihood still improves, even though the curve looks almost like a step.

Slide 8 - notes and edits

Likelihood Along This Sequence of Curves



The log likelihood approaches 0 as $\beta \to \infty$. No finite slope reaches the supremum.

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An Infinite Maximum Likelihood Estimate

For these individual responses,

$$L(\alpha, \beta) = \prod_i \pi_i^{y_i}(1 - \pi_i)^{1-y_i}.$$

Along $\alpha = -50\beta$, increasing β makes every factor closer to 1.

- The likelihood approaches 1, but every finite logistic curve has probabilities strictly between 0 and 1.
- There is no finite maximizer: we describe the slope estimate as $\hat{\beta} = \infty$.
- The intercept also diverges along this path.

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Fitting the Model in R

```
> fit <- glm(y ~ x, family = binomial, data = toy)
Warning: glm.fit: fitted probabilities numerically 0 or 1
    occurred
```

```
> summary(fit)
Call:
glm(formula = y ~ x, family = binomial, data = toy)
Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)   -118.16  296046.19       0        1
x                2.36    5805.94       0        1
(Dispersion parameter for binomial family taken to be 1)
    Null deviance: 11.09035488895912  on 7  degrees of freedom
Residual deviance:  0.00000000021827  on 6  degrees of freedom
AIC: 4
Number of Fisher Scoring iterations: 25
```


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The Wald Test

The reported test for the slope uses

$$z = \frac{\hat{\beta}}{\widehat{\text{SE}}(\hat{\beta})} \approx \frac{2.36}{5805.94} \approx 0.$$

- R stops at a finite numerical value, even though the ML estimate is infinite.
- The reported standard error is enormous.
- The resulting Wald p-value is approximately 1.

Does $p \approx 1$ make sense when x perfectly separates the outcomes?

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A Likelihood-Ratio Test

Compare the intercept-only model to the model with x:

```
> fit_null <- glm(y ~ 1, family = binomial, data = toy)
> anova(fit_null, fit, test = "LRT")
Analysis of Deviance Table
Model 1: y ~ 1
Model 2: y ~ x
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1         7       11.1
2         6        0.0  1     11.1  0.00087 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The deviance decrease is 11.09 on one degree of freedom, giving an approximate $p = 0.0009$. The likelihood-ratio test provides evidence of an association even though the Wald test fails. Its chi-square reference is a large-sample approximation.

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Quasi-Complete Separation

Now add two observations at the cutoff:

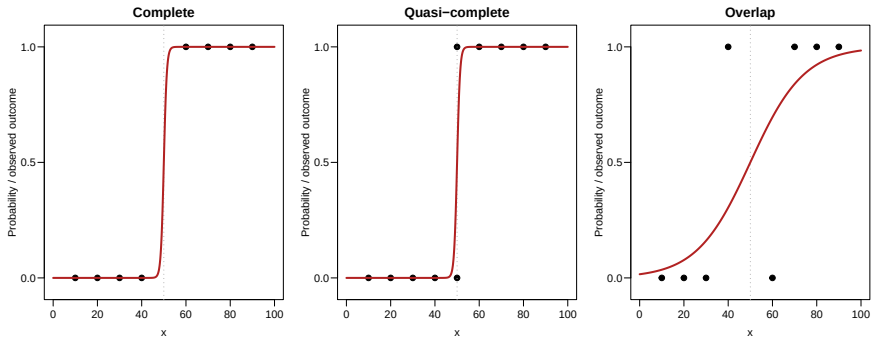
$$(x, y) = (50, 0) \quad \text{and} \quad (50, 1).$$

- Away from the cutoff, the outcomes still separate.
- Both outcomes occur at $x = 50$.
- Along $\alpha = -50\beta$, both cutoff observations keep fitted probability 0.5.
- The other observations are fit increasingly well as β increases.

The slope estimate is still infinite, but the limiting likelihood is below 1.

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Separation and Overlap



Complete and quasi-complete panels show a steep finite curve approaching the limit. The overlap panel shows an ordinary finite ML fit. For several predictors, a linear combination of the predictors may separate the outcomes even when no single predictor does so alone.

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Separation with a Categorical Predictor

	$Y = 1$	$Y = 0$
$X = 1$	15	0
$X = 0$	12	8

$$\widehat{\text{OR}} = \frac{15 \times 8}{0 \times 12} = \infty, \qquad \hat{\beta} = \log \widehat{\text{OR}} = \infty.$$

- Both outcomes occur for $X = 0$, but only successes occur for $X = 1$.
- This is quasi-complete separation.

Illustrative counts. In larger tables, a zero cell need not imply separation for every model.

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Endometrial Cancer Data

For 79 patients, HG is histology grade (0 = low, 1 = high). NV indicates neovasculation (0 = absent, 1 = present).

```
> Endo <- read.table("data/Endometrial.dat", header = TRUE)
> head(Endo)
  NV PI   EH HG
1  0 13 1.64  0
2  0 16 2.26  0
3  0  8 3.14  0
4  0 34 2.68  0
5  0 20 1.28  0
6  0  5 2.31  0
> xtabs(~ NV + HG, data = Endo)
   HG
NV   0  1
  0 49 17
  1  0 13
```

All 13 patients with NV = 1 have HG = 1. Which type of separation is this?

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A Model for Histology Grade

The model also includes PI (pulsatility index) and EH (endometrium height):

$$\text{logit}\{P(\text{HG} = 1)\} = \beta_0 + \beta_{\text{NV}}\text{NV} + \beta_{\text{PI}}\text{PI} + \beta_{\text{EH}}\text{EH}.$$

- Increasing β_{NV} improves predictions for the NV = 1 patients.
- It leaves the predictions for NV = 0 patients unchanged when the other coefficients are held fixed.
- Hence $\hat{\beta}_{\text{NV}} = \infty$.

What do you expect the coefficient table to look like?

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Endometrial Cancer Model in R

```
> endo_fit <- glm(HG ~ NV + PI + EH, family = binomial,
+                 data = Endo)
> summary(endo_fit)
Call:
glm(formula = HG ~ NV + PI + EH, family = binomial, data = Endo)
Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)    4.3045     1.6373    2.63   0.0086 **
NV            18.1856  1715.7509    0.01   0.9915
PI            -0.0422     0.0443   -0.95   0.3413
EH            -2.9026     0.8456   -3.43   0.0006 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
(Dispersion parameter for binomial family taken to be 1)
    Null deviance: 104.903  on 78  degrees of freedom
Residual deviance:  55.393  on 75  degrees of freedom
AIC: 63.39
Number of Fisher Scoring iterations: 17
```


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Testing the NV Effect

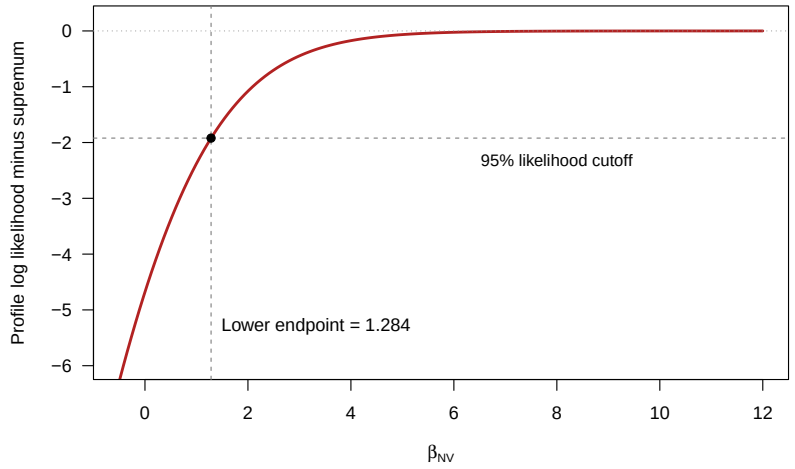
Compare models with and without NV, keeping PI and EH:

```
> endo_without_NV <- update(endo_fit, . ~ . - NV)
> anova(endo_without_NV, endo_fit, test = "LRT")
Analysis of Deviance Table
Model 1: HG ~ PI + EH
Model 2: HG ~ NV + PI + EH
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1        76       64.8
2        75       55.4  1     9.36   0.0022 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The Wald p-value for NV was 0.9915. The likelihood-ratio p-value is about 0.0022. An infinite ML estimate does not prevent likelihood-based inference about the effect.

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Profile Likelihood for the NV Coefficient



For each fixed β_{NV}, refit the other coefficients. The profile approaches its supremum as $\beta_{\text{NV}} \to \infty$.

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A Profile Likelihood Confidence Interval

Values satisfying

$$2\{\ell_{\text{sup}} - \ell_{\text{profile}}(\beta_{\text{NV}})\} \leq \chi^2_{1,0.95} \approx 3.84$$

form an approximate 95% likelihood confidence interval.

```
> c(lower = ci_lower, upper = Inf)
 lower  upper
1.2841    Inf
> exp(c(lower = ci_lower, upper = Inf))
 lower  upper
3.6115    Inf
```

The interval for β_{NV} is about $(1.28, \infty)$, corresponding to an odds-ratio interval of about $(3.61, \infty)$, adjusting for PI and EH.

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Recognizing Infinite Estimates

- Examine outcome counts within predictor categories and the arrangement of outcomes along numerical predictors.
- Watch for very large coefficients and standard errors, fitted probabilities close to 0 or 1, and fitting warnings.
- Numerical convergence can occur without a finite ML estimate.

A sample with only one outcome in a group does not establish that the population probability is exactly 0 or 1.

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Inference and Finite Estimates

- Wald inference can be misleading when ML estimates are infinite.
- Likelihood-ratio tests and profile likelihood intervals can remain informative.
- Penalized likelihood and Bayesian methods can produce finite estimates under separation, using suitable penalties or priors.

Agresti introduces alternatives to ordinary maximum likelihood in Section 5.4.

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Review

1. Why does increasing β keep improving the likelihood for the toy data?
2. Why do we also change α in the curve sequence?
3. How does quasi-complete separation differ from complete separation?
4. Why can a Wald test report $p \approx 1$ for a predictor that perfectly separates outcomes?
5. What can we conclude from a profile interval with an infinite upper endpoint?

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Appendix: The Sequence of Curves in R

```
> beta <- c(0.02, 0.10, 0.50, 1.50)
> alpha <- -50 * beta
> log_likelihood <- sapply(beta, function(b) {
+   sum(dbinom(y, size = 1, prob = plogis(b * (x - 50)),
+              log = TRUE))
+ })
> data.frame(beta, alpha, log_likelihood)
  beta alpha log_likelihood
1 0.02    -1  -3.8394854764
2 0.10    -5  -1.0138539561
3 0.50   -25  -0.0135221107
4 1.50   -75  -0.0000006118
```

This sequence demonstrates an improving path through parameter space. The intercept changes with the slope to keep the cutoff at 50.

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Appendix: Numerical Convergence

```
> c(iterations = fit$iter, converged = fit$converged)
iterations  converged
        25          1
> logLik(fit)
'log Lik.' -0.00000000010913 (df=2)
```

The algorithm reports convergence at a finite numerical approximation. The mathematical maximum is not attained at finite coefficients.

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Sources

- Agresti, *An Introduction to Categorical Data Analysis*, third edition, Section 5.3, pp. 136–140.
- Toy observations: Figure 5.2 and Section 5.3.2.
- Endometrial cancer data: Table 5.4 and Section 5.3.4.
- Original data: Alan Agresti's public data repository.

The toy-data curves and likelihood path, overlap illustration, categorical table, and reproduced endometrial profile are lecture illustrations computed from the models described here.

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